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# Interferon-stimulated gene *GALNT2* restricts respiratory virus infections

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## **Supplementary methods**

### **Expression and regulation of genes across tissues and COVID-19 severity groups**

GALNT2 expression patterns were obtained online from the Human Protein Atlas (<https://www.proteinatlas.org>). Transcriptomic expression levels were z-score normalized across tissues and visualized using heatmaps to highlight relative abundance. Pairwise correlations among OAS1, MX1, and GALNT2 transcript levels were assessed using Spearman's rank correlation and visualized with scatter plots fitted with locally estimated scatterplot smoothing (LOESS) lines and 95% confidence intervals.

Protein-level data were categorized into four expression levels—"not detected", "low", "medium", and "high"—according to database annotations. These classifications were summarized across tissues in a categorical heatmap.

To evaluate gene regulation in the context of SARS-CoV-2 infection, we obtained proteomics data from a published COVID-19 study<sup>46</sup>. Differential protein expression between SARS-CoV-2 infected patients and non-COVID controls across seven organs was extracted from the supplementary tables. Proteins with  $|\log_2(\text{fold change})| > \log_2(1.2)$  and Benjamini–Hochberg-adjusted  $p < 0.05$  were considered significantly differentially expressed.

### **Immunofluorescence assay (IFA)**

Infected cells were fixed with 4% PFA. Cells were washed with PBS, and then permeabilized with 0.2% Triton-X 100 in PBS within 1%BSA for 30 minutes. After permeabilization, cells were washed 3 times with PBS and then primary antibody (rabbit anti-SARS-CoV-2 nucleocapsid protein polyclonal antibody (Sino Biological, 40143-T62, Beijing, China), rabbit anti-HCoV-OC43 nucleocapsid protein polyclonal antibody (Sino Biological, 40643-T62, Beijing, China), rabbit anti-HCoV-229E nucleocapsid protein polyclonal antibody (Sino Biological, 40640-T62, Beijing, China) and mouse anti-pan Influenza A nucleoprotein monoclonal antibody (Sino Biological, 40208-R010, Beijing, China) was added in blocking

solution and incubated for 1 hour. Cells were washed 3 times with PBS, after which secondary antibody [Alexa Fluor® 488 AffiniPure Donkey Anti-Rabbit IgG (H+L) (Jackson ImmunoResearch Laboratories, 711-545-152, Inc. West Grove, PA), Alexa Fluor® 488 AffiniPure Goat Anti-Mouse IgG (H+L) (Jackson ImmunoResearch Laboratories, 115-545-003, Inc. West Grove, USA) was added in 1% BSA and incubated for 1 hour. Cells were washed 3 times with PBS, and then stained with DAPI (Beyotime, Shanghai, China) for 10 minutes (1:5000, Beyotime, Shanghai, China). Images of the cells were taken with the Nexcelom Celigo Image Cytometer (Nexcelom Bioscience, Inc. Boston, USA). All primary antibodies were used at a dilution of 1:2,000 for immunofluorescence staining. All secondary antibodies were used at a dilution of 1:1,500 for immunofluorescence.

### **RNA-sequencing and data analysis**

Total RNA was extracted from lungs of infected mice or cells using Trizol (Invitrogen, Carlsbad, USA) according to the manufacturer's protocol, and ribosomal RNA was removed using QIAseq FastSelect-rRNA HMR Kits (QIAGEN, Germany). Fragmented RNA (average length approximately 200 bp) was subjected to first strand and second strand cDNA synthesis followed by adaptor ligation and enrichment with a low-cycle according to the instructions of NEBNext Ultra™ RNA Library Prep Kit for Illumina (NEB, Ipswich, USA). The purified library products were evaluated using the Agilent 2200 TapeStation and Qubit2.0 (Life Technologies, Carlsbad, USA). The libraries were paired-end sequenced (PE150, Sequencing reads were 150 bp) at Guangzhou RiboBio Co., Ltd. (Guangzhou, China) using Illumina HiSeq 3000 platform (Illumina HiSeq 3000, Inc. San Diego, USA). Data analysis method has reported in our previous study<sup>13</sup>. Briefly, reads were aligned to the mouse or human reference genome mm10 with bowtie2 and RSEM were used to quantify the reads mapped to each gene. Gene expression was normalized by EDASEQ and differentially expressed genes were obtained using DESeq2 (version 1.10.1), with a cutoff of Q-value < 0.05 and log2 (fold-change) >1. All

differentially expressed mRNAs were selected for GO analyses clusterProfiler and deposited to the Gene Expression Omnibus (GEO) database: GSE301783.

## **SILAC-based quantitative proteomics and glycoproteomics**

A549 cells were grown in SILAC DMEM (Thermo Fisher) supplemented with L-arginine and L-lysine (Sigma-Aldrich), while the heavy isotope counterparts L-<sup>13</sup>C<sub>6</sub>, <sup>15</sup>N<sub>4</sub>-arginine and L-<sup>13</sup>C<sub>6</sub>, <sup>15</sup>N<sub>2</sub>-lysine (Sigma-Aldrich) were employed for A549 GALNT2-OE cells. Arginine and lysine were added at concentrations of 84 and 146 mg/L, respectively. The SILAC media were further supplemented with 10% FBS, penicillin (100 U/mL), streptomycin (100 mg/mL), 1% GlutaMax (Thermo Fisher), and 10 mM HEPES (pH 7.5) (Corning). After growing for 12 doublings, the cells were lysed using 0.4% Rapigest in 50 mM Ammonium Bicarbonate and the extracted protein was quantified using the BCA assay and mixed in a 1:1 ratio. The mixed proteins were reduced with 10 mM DTT for 45 min at 60 °C and then alkylated with 20 mM iodoacetamide for 30 min at room temperature in the dark. The reaction was then quenched with 10 mM DTT in the dark for an additional 30 minutes. The proteins were then digested with trypsin (Promega) in a protein: enzyme ratio of 1:50 at 37 °C overnight. Following digestion, neuraminidase (Sigma-Aldrich) was added to remove the sialic acid. Then the reaction mixture was cleaned up by a Sep-Pak Cartridge (Waters) and an aliquot of peptides were collected for proteomics. For glycoproteomics, the remaining purified peptides were lyophilized and dissolved in binding buffers (175 mM Tris-HCl, pH 7.4), then subjected to glycopeptide enrichment by Jacalin agarose (Vector) as described in literature<sup>50</sup>. The enriched glycopeptides were then purified and desalted again by Stage-Tip for LC-MS/MS analysis.

The SILAC-labeled peptides were analyzed on Q Exactive Plus with a nano-electrospray source. The SILAC-labeled glycopeptide products were purified and desalted by Stage-Tip, and analyzed using an U3000nanoRSLC coupled to an Eclipse Mass Spectrometer equipped

with a Nanospray Flex Ion Source (Thermo Fisher). The samples dissolved in 0.1% FA were separated on a single analytical column (Acclaim PepMap 100 C18, 2 $\mu$ m, 75 $\mu$ m inner diameter, 25cm length) at a flow rate of 300 nL/min over a 120-min gradient (buffer A, 0.1% FA in water; buffer B, 0.1% FA in 80% ACN) of 1-5% B for 3 min, 5-30% B for 110 min, 30-90% B for 4 min, 90% B for 3 min. The mass spectrometer was set to acquire full scan MS spectra (300-1500m/z) for a maximum injection time of 100ms at a mass resolution of 1.2e5 and an automated gain control (AGC) target value of 4e5. The MS2 analyses were performed using data dependent acquisition with dynamic exclusion set to 30s at exclusion window of 10ppm, and the cycle time set to 3s. During every cycle time, the most abundant multiply charged precursors from the full scan were selected for fragmentation via electron-transfer/collision-induced dissociation (ETciD) in the orbitrap at 5e4 resolution. The calibrated charge dependent ETD parameters were set as true. The AGC target was set to 1.5e5 and the maximum injection time was 86 ms. Data were searched against the UniProtKB/Swiss-Prot Homo sapiens database using SequestHT as searching engine through Thermo Scientific Proteome Discoverer (v 2.1). Enzyme restriction was set to trypsin digestion with full specificity (N-terminal to lysine and arginine) and a maximum of five missed cleavages. The mass tolerances in MS and MS/MS were set to 10 ppm and 0.02 Da, respectively. Carbamidomethylation (C) was used as a fixed modification, and oxidation (M) was used as a variable modification. For glycoproteomics, additional variable modifications include HexNAc (S,T), HexNAc(2) (S,T), Hex(1)HexNAc(1) (S,T) , Hex(1)HexNAc(2) (S,T). The FDR threshold was set at 1% at both peptide and protein levels in the Peptide and Protein FDR Validator node. Three replicates were included for SILAC quantitation using the Minora Feature Detector Node and SILAC 2plex as quantification methods. ANOVA test was used to estimate the ratio significance. Glycosylated peptides that met both thresholds (adjusted  $P \leq 0.05$  and fold change  $\geq 2$ ) were defined as significantly regulated.

## **Reagents and plasmids**

Rabbit anti-S2 (Sino Biological, 40590-T62) antibody was used for the detection of S2', S2 and S proteins. ACE2 polyclonal antibody (Proteintech, 21115-1-AP), anti-GAPDH-HRP (Abclonal, AC035), HRP conjugated goat anti-mouse (Proteintech, SA00001-1) and anti-rabbit (Proteintech, SA00001-2) secondary antibodies were used for immunoblot. Codon-optimized (for Homo sapiens) genes encoding SARS-CoV-2 spike proteins (WT and Omicron BA.4/5) and other plasmids were synthesized and cloned into the pcDNA3.1 vector. Site-directed mutagenesis and deletions were introduced according to the manufacturing protocol.

All primary antibodies were used at a dilution of 1:1000 for immunoblotting, 1:100 for immunoprecipitation, 1:2000 for immunofluorescence staining and 1:2000 for focus forming assay. All secondary antibodies were used at a dilution of 1:2500 or 1:5000 for immunoblotting, 1:1000 for immunofluorescence staining and 1:5000 for focus forming assay.

### **Transient transfection**

Huh7, Caco-2-N, Calu3 and HEK293T cells were transfected with Lipofectamine 3000 (Thermo Fisher Scientific) according to the manufacturer protocols. The GALNT2 transfection efficiency was confirmed by WB, IFA and FACS assays, as well as the control plasmids (EV and eGFP) prior to functional assays.

CellTiter-Glo® luminescent cell viability assay (#G7570, Promega AG) was used to assess viability dependent on ATP availability according to the manufacturer's protocol in a 96-well format. Bioluminescence was determined using a BioTek Synergy H1. Relative cell growth was determined daily and normalized to WT cells.

### **Lentiviral pseudoparticle production and transduction**

All lentiviral pseudoparticles were generated by co-transfected 293T cells with expression plasmids [pLentiCRISPRv1. GALNT2/STAT1. puromycin or pLOC. CMV. IRSE. GALNT2. blasticidin], HIV-1 gag-pol, VSV-glycoprotein at a ratio of 2:2:1 using PEI40000. Four to six

hours post transfection, media was replaced with fresh DMEM containing 10% FBS. Supernatants were collected at 60 h.p.t., pooled, aliquoted, and stored at -80°C. The following target sequences for the guides were used: GALNT2 guides (guide 1: TGAACCTTGTTGCGGGCGTAA, guide 2: CGCCCGCAACAAGTTCAACC), STAT1 guides (guide 1: TGTGATAGGGTCATGTTCGT, guide 2: GATCACCACAACGGGCAGAG).

Cells were transduced by spinoculations. Briefly, Huh7-Cas9 or A549-Cas9 cells were seeded in 24-well plates, lentivirus was added to a minimum volume of transduction media (10% FBS, appropriate base media for each cell line, 8 µg/mL polybrene) before adding to cells. Cells were centrifuged at 800 g at 33°C for 2h, and then media was replaced with standard growth media. Huh7 GALNT2<sup>-/-</sup> and Huh7 STAT1<sup>-/-</sup> cell lines were selected for 5-7 days in the presence of 1 µg/mL puromycin. GALNT2 overexpression in Huh7 and A549 cells (Huh7 OE-GALNT2 and A549 OE-GALNT2) were selected for 7-10 days in 5 µg/mL blasticidin. GALNT2<sup>-/-</sup>, STAT1<sup>-/-</sup>, Huh7 OE-GALNT2 and A549 OE-GALNT2 cell lines were validated by western blot.

Huh7 OE-GALNT2 and A549 OE-GALNT2 cell lines were maintained in the presence of 5 µg/mL blasticidin (InvivoGen, France). GALNT2<sup>-/-</sup> in Huh7 and A549 cells were maintained in the presence of 0.75 µg/mL puromycin (InvivoGen, France)

### **Immunoblotting**

Syncytia and cell mixtures were lysed directly in 2 x Laemmli buffer, boiled at 95 °C for 5 min, and separated via 10% Tris-glycine SDS-PAGE. Proteins were transferred to PVDF membranes, blocked with 2.5% BSA in PBST, and probed with primary antibodies (1 h, RT). HRP-conjugated secondary antibodies and ECL substrate were used for detection, with imaging performed on a Tanon 5200 system. All primary antibodies were used at a dilution of 1:100 for immunoblotting. All secondary antibodies were used at a dilution of 1:2,500 or 1:5000 for immunoblotting.

## **Assessment of plasmids transfection efficiency by flow cytometry (FACS)**

Huh7 and HEK293T cells were seeded in 24-well plates overnight before transiently transfected with 300 ng plasmids of empty vector (EV), GFP or GALNT2-HA packaged in Lipofectamine 3000 (Life technologies). 24 hours post-transfection, HEK293T and Huh7 cells were detached using trypsin-EDTA and fixed in 4% PFA. For staining of the GALNT2-HA tag, cells were further permeabilized in 0.1% Triton-X100 and blocked in 1% BSA PBS. APC-conjugated mouse Anti-HA tag antibody (901523, Biolegend) was then added at 1:200 dilution to cell suspension for the staining of GALNT2 for 1 hour. Flow cytometry was carried out on a NovoCyte Advanteon (Agilent), subsequent gating and data analysis was performed using the FlowJo 10.4 (LCC).

## **Expression and purification of GALNTs from HEK293T cells**

The encoding gene of GALNTs without N terminal transmembrane domain was cloned into the pSec-Tag2A expression vector, and plasmids were transfected into FreeStyle™ HEK293T cells using linear polyethyleneimine transfection reagent (PolySciences, Chicago, USA). The secreted GalNAc-Ts proteins were purified with Ni-NTA column from supernatant. The purified proteins were analyzed by SDS-PAGE (10% Tris-glycine gel) and Coomassie staining.

## **Transduction and infection of mice**

Recombinant adenovirus vector expressing hACE2 (Ad5-hACE2) was prepared and used as previously described<sup>13</sup>. The Ad5-hACE2 or Ad5-Empty vectors are available from the University of Iowa. AAV9-EV and AAV9-GALNT2 stocks were purchased from Packgene Biotech Co and transduced intranasally with  $3.3 \times 10^{11}$  gene copies in mice. Nine days later, mice were anesthetized with isoflurane and transduced intranasally with  $2.5 \times 10^8$  FFU of Ad5-hACE2. Five days post transduction, lungs of transduced mice were harvested and GALNT2 expression levels were determined by western blot and qPCR assays. Mice were

infected intranasally with  $1 \times 10^5$  PFU SARS-CoV-2 (WT) or 350 to 1000 FFU H1N1 PR8 in a total volume of 50 $\mu$ L DMEM, respectively.

## **Histopathology**

Mice, after cardiac perfusion as described above, were immersion-fixed with 10% neutral-buffered formalin solution. Tissue was embedded in paraffin, including the whole lung, three to four trimmed sections of a sagittal section of the lung. Samples were cut into 2-3 $\mu$ m thick sections and stained with H&E for light microscopical examination.

## Supplementary Materials

### Extended Data Figure 1 to 10

Extended Data Fig. 1. Impaired type I IFN response leads to increased and prolonged SARS-CoV-2 infection *in vivo*

Extended Data Fig. 2. Single-cell transcriptional profiling of BALF and PBMCs from HDs and SARS-CoV-2 infected patients.

Extended Data Fig. 3. Expression and regulation of GALNT2 and classical interferon-stimulated genes across tissues and COVID-19 severity groups.

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Extended Data Fig. 6. Increased lethality and pathogenicity of H1N1 PR8 in *GALNT2*<sup>-/-</sup> mice.

Extended Data Fig. 7. Overexpression of GALNT2 alleviates pneumonia induced by IAV and SARS-CoV-2 infection.

Extended Data Fig. 8. Overexpression of GALNT2 in A549 has a minimal impact on host genes.

Extended Data Fig. 9. GALNT2-dependent O-linked glycosylation at S810 and S813 of spike.

Extended Data Fig. 10. The S810/813I evades GALNT2 antiviral function in a SARS-CoV-2 replicon system.

**Extended Data Fig. 1. Impaired type I IFN response leads to increased and prolonged SARS-CoV-2 infection *in vivo*.**

**a**, A schematic showing the mouse infection model of SARS-CoV-2 by exogenous delivery of human ACE2 with a replication-deficient adenovirus (Ad5-hACE2). **b**, *IFNAR*<sup>-/-</sup> (n=3) and WT (n=3) were intranasally transduced with 2.5×10<sup>8</sup> FFU of Ad5-hACE2. Five days post transduction of Ad5-hACE2, mice were infected with 1×10<sup>5</sup> PFU of SARS-CoV-2 (WT). The lungs were harvested at 2 days post infection and performed mRNA-seq. The transcriptional profiling of homogenized lungs are expressed as normalized read counts to WT mice. **c**, The genes showed in (**b**) were homo sapiens codon-optimized and synthesized *de novo* into pcDNA3.1 vector. Huh7 cells were transfected with plasmids encoding these genes (EV and eGFP as the controls). At 24h post transfection, cells were infected with SARS-CoV-2 (WT, MOI=0.05) for 48 h. Viral infection efficiency was determined by IFA. Data were normalized to EV. Data are means ± s.d. from n = 3 independent experiments. **d**, Huh7 cells were transfected with plasmid encoding GALNT2 (EV as the control), and GALNT2 protein expression level was detected at 24 hours post transfection by western blots. Representative IFA images that GALNT2 inhibits SARS-CoV-2 infection in Huh7 cells. SARS-CoV-2 N was stained using indirect fluorescent antibody (green), nuclei are stained with DAPI (blue). Western blots and IFA images were representative of three independent biological repeats. **e**, GALNT2 and STAT1 protein levels of WT and *IFNAR*<sup>-/-</sup> mice (n=3) at 2, 4, and 6 days post infection were determined by western blots. **f**, Related to (**e**), comparison viral titers and GALNT2 protein levels of WT mice at 2, 4, and 6 d.p.i.. n=3 mice per group. GALNT2 protein levels were normalized to that in WT mice (4 d.p.i.). **g**, Related to (**e**), comparison viral titers and GALNT2 protein levels of *IFNAR*<sup>-/-</sup> mice at 2, 4 and 6 d.p.i.. n=3 mice per group. GALNT2 protein levels were normalized to that in WT mice (4 d.p.i.). **h**, Huh7 cells were transfected with plasmids encoding GALNT2-HA (EV as the control) for 36 hours. Cells were then fixed with 4% paraformaldehyde and stained using Anti-HA-tag antibodies conjugated with Allophycocyanin (APC). Cells were analyzed by IFA. Scale bars are indicative of 20 μm. **i**,

244 Huh7 and HEK293T cells were transfected with plasmids encoding EV, eGFP and  
245 GALNT2-HA for 24 hours. GALNT2 expression levels were detected by flow cytometry and  
246 data analysis was performed using the FlowJo 10.4 (LCC). Data are means  $\pm$  s.d. from  $n = 3$   
247 independent experiments. **j**, Huh7 cells were transfected with plasmids encoding GALNT2  
248 (EV and eGFP as the controls. At 24h post transfection, the cell viability was measured by  
249 CCK-8 and normalized to Huh7 WT cells. Data are means  $\pm$  s.d. from  $n = 12$  independent  
250 experiments. **k**, Huh7 cells were transfected with plasmids encoding GALNT2 (EV and eGFP  
251 as the controls. At 24h post transfection, cells were seeded in 24-well plates at  $1 \times 10^4$   
252 cells/well. Cell growth is counted at indicated time points. Data are means  $\pm$  s.d. from  $n = 3$   
253 independent experiments. *P* values from two-tailed Student's t-test.

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**Extended Data Fig. 2. Single-cell transcriptional profiling of BALF and PBMCs from HDs and SARS-CoV-2 infected patients.**

**a**, Canonical markers for different cell types in BALF were shown in UMAP and dot plots. **b**, The UMAP plots were split by different studies. **c**, Comparison of GALNT2 expression in BALF samples using scRNA-seq. Each dot represents an individual, and the y-axis represents the average expression level of GALNT2 in a specific cell type from that individual. **d**, Comparison of GALNT2 expression in Myeloid, T and B, ciliated and secretory cells of BALF samples using scRNA-seq. Each dot represents an individual, and the y-axis represents the average expression level of GALNT2 across all cells or specifically in ciliated cells from that individual. The box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the whiskers extend to the smallest and largest values within 1.5 times the interquartile range from the box. The *p* values were calculated by two-tailed Wilcoxon signed-rank test. ((ns): no significant,  $*P \leq 0.05$ ,  $**P \leq 0.01$ ,  $***P \leq 0.001$ ,  $****P \leq 0.0001$ ). **e**, Protein expression of GALNT2 in the PBMC samples at conversion phase (T2) from mild and severe patients were detected by western blots. Western blots were representative of three independent biological repeats. Quantitative data from representative western blot is shown. Data are means  $\pm$  s.d. from  $n = 3$  independent experiments. *P* values from two-tailed Student's t-test. ((ns): no significant,  $*P \leq 0.05$ ,  $**P \leq 0.01$ ,  $***P \leq 0.001$ ,  $****P \leq 0.0001$ ).

**Extended Data Fig. 3. Expression and regulation of GALNT2 and classical ISGs across tissues and COVID-19 severity groups.**

**a**, RNA expression patterns of OAS1, MX1, and GALNT2 across human tissues (data from the Human Protein Atlas; <https://www.proteinatlas.org>), visualized as z-score normalized heatmaps (red: higher expression, blue: lower expression). Scatter plots below show pairwise expression correlations between genes, with locally estimated scatterplot smoothing (LOESS) fits (blue line) and 95% confidence intervals (gray shading). Pearson correlation coefficients and p-values by two-tailed test are indicated. **b**, Protein-level expression of OAS1, MX1, and GALNT2 across human tissues, from the Human Protein Atlas. Expression levels are categorized as not detected (grey), low (light blue), medium (blue), or high (dark blue) by HPA database. **c**, Left: Comparison of OAS1 expression in ciliated epithelial cells from bronchoalveolar lavage fluid (BALF) across COVID-19 severity groups based on scRNA-seq data. The box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the whiskers extend to the smallest and largest values within 1.5 times the interquartile range from the box. Each dot represents an individual; the y-axis shows average OAS1 expression per individual. Right: Longitudinal OAS1 expression in peripheral blood mononuclear cell (PBMC) subsets (B cells, T cells, and monocytes) across healthy controls (HC) and patients with mild or severe COVID-19. Open dots represent individual-level averages; solid dots represent group means. Time points: T1 (acute phase), T2 (conversion), T3 (recovery; see Methods for definitions). Samples from the same individual are connected by dashed lines. P values were calculated using two-tailed Wilcoxon signed-rank test (left) or two-tailed Student's t-test (right). **d**, Comparison of MX1, SOCS1, STAT1, and USP18 expression in ciliated BALF cells across clinical severity groups. Each dot represents one individual; the y-axis shows average gene expression. The box plots show the median (center line), the 25th and 75th percentiles (bounds of the box), and the whiskers extend to the smallest and largest values within 1.5 times the interquartile range from the box. P values by two-tailed Wilcoxon signed-rank test. **e**, Summary of differential protein expression for the

302 indicated genes across tissues in SARS-CoV-2 infected patients versus controls, based on  
303 published proteomics data. Red indicates significant upregulation (FDR < 0.05); gray, not  
304 significant. ((ns): no significant, \* $P \leq 0.05$ , \*\* $P \leq 0.01$ , \*\*\* $P \leq 0.001$ , \*\*\*\* $P \leq 0.0001$ ).

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**Extended Data Fig. 4. GALNT2 associated with the severity or susceptibility of SARS-CoV-2 and IAV infections.**

**a**, GALNT2 association results in influenza A virus H7N9 genome-wide association studies (GWAS). Mx1 gene (left panel) as the control. Red plots represent the SNPs associated with virus infection. **b**, GALNT2 association results in influenza A virus H1N1 genome-wide association studies (GWAS). Mx1 gene (left panel) as the control. Red plots represent the SNPs associated with virus infection. *P* values from **(a)** and **(b)** were calculated by two-tailed Firth logistic regression test, with appropriate covariates and genomic control. **c**, Analysis of genetic variation at the GALNT2 gene locus between SARS-CoV-2 infected patients and control populations using data from Host Genetics Initiative (<https://www.covid19hg.org/>). Population frequency information for the SNP was sourced from <https://www.ncbi.nlm.nih.gov/snp/rs2273970>. *P* values were calculated by two-tailed Firth logistic regression test, with appropriate covariates and genomic control. **d**, Huh7 cells were transfected with plasmids encoding GALNT2 (WT or four mutants), and cells then infected with SARS-CoV-2 (MOI=0.5) and viral infection efficiency after 24 h.p.i. by IFA. Data are means  $\pm$  s.d. from *n* = 3 or 5 independent experiments. *P* values from two-tailed Student's *t*-test. ((ns): no significant, \**P*  $\leq$  0.05, \*\**P*  $\leq$  0.01, \*\*\**P*  $\leq$  0.001, \*\*\*\**P*  $\leq$  0.0001). **e**, Huh7 cells were transfected with plasmids encoding GALNT2 (WT or V554M), and cells then infected with H1N1 PR8 (MOI=0.01) and viral infection efficiency after 24 h.p.i. by IFA. Data are means  $\pm$  s.d. from *n* = 3 independent experiments. *P* values from two-tailed Student's *t*-test. ((ns): no significant, \**P*  $\leq$  0.05, \*\**P*  $\leq$  0.01, \*\*\**P*  $\leq$  0.001, \*\*\*\**P*  $\leq$  0.0001). **f**, GALNT2 protein expression levels in Huh7 OE-GALNT2 (GALNT2-stable cells) and Huh7 GALNT2<sup>-/-</sup> cells were detected by western blot. Western blots were representative of three independent biological repeats. **g**, GALNT2 protein expression levels in A549 OE-GALNT2 (GALNT2-stable cells) and A549 GALNT2<sup>-/-</sup> cells were detected by western blot. Western blots were representative of three independent biological repeats. **h**, STAT1 protein expression levels in Huh7 STAT1<sup>-/-</sup> and A549 STAT1<sup>-/-</sup> cells were detected by western blot.

Western blots were representative of three independent biological repeats. **i**, Huh7 WT, GALNT2<sup>-/-</sup> and reconstitution of GALNT2 in the knockout cells by transient transfection (eGFP as the control) were infected with HCoV-OC43 at MOI 0.05. Supernatants were harvested at 24, 48 and 72 h.p.i. and viral titers were determined in HRT-18 cells by FFA. Data are means  $\pm$  s.d. from n = 3 independent experiments. **j**, Huh7 WT, GALNT2<sup>-/-</sup> and reconstitution of GALNT2 in the knockout cells by transient transfection (eGFP as the control) were infected with HCoV-229E at MOI 0.001. Supernatants were harvested at 24, 48 and 72 h.p.i. and viral titers were determined in Huh7 cells by FFA. Data are means  $\pm$  s.d. from n = 3 independent experiments. **k**, A549 WT (Ctrl) and A549 GALNT2<sup>-/-</sup> cells were infected with IAVs at MOI of 0.01, including H1N1 (PR8 and CA04) and H3N2 (HK68). Supernatants were harvested at indicated time points and viral titers were determined on MDCK cells by FFA. Data are means  $\pm$  s.d. from n = 3 independent experiments. Data are means  $\pm$  s.d. from n = 3 independent experiments. *P* values from two-tailed Student's t-test. (ns): no significant, \**P*  $\leq$  0.05, \*\**P*  $\leq$  0.01, \*\*\**P*  $\leq$  0.001, \*\*\*\**P*  $\leq$  0.0001).

**Extended Data Fig. 5. GALNT2 is an antiviral ISG.**

**a-b**, AAV9-EV and AAV9-GALNT2 transduced Calu3 cells. At 48h post transduction, cells were infected with SARS-CoV-2 WT at MOI 0.01, supernatants were harvested at 48 h.p.i. and viral titers were determined in Vero E6 cells by FFA. Images of viral foci were captured by ELISPOT. *P* values from two-tailed Student's t-test. **c-d**, Foci forming sizes (n=20) were measured and analyzed. **e**, Quantative PCR analysis of GALNT2 mRNA in Huh7 cells after SARS-CoV-2 infection at MOI of 0.5. Data are means  $\pm$  s.d. from n = 3 independent experiments. **f**, THP-1 cells were infected HCoV-OC43 at MOI of 1. The mRNA expression levels of GALNT2 was analyzed by qPCR at indicated time points. Data are means  $\pm$  s.d. from n = 3 independent experiments. **g**, Huh7 cells were infected with HCoV-OC43 for 6 (n=3) or 24h (n=3), respectively. GALNT2 and phosphorylated STAT1 (pSTAT1) protein expression levels were detected by western blot. **h**, A549 cells were infected with PR8 H1N1 at MOI of 0.01. Cell lysates were harvested at 6 and 24 h.p.i., respectively. GALNT2 and pSTAT1 protein expression levels were detected by western blot. Western blots were representative of three independent biological repeats. **i**, Huh7 STAT1<sup>-/-</sup> and Huh7 cells (Ctrl) were treated with 5  $\mu$ g/mL of IFN- $\alpha$ 2 for 24h. The mRNA levels of GALNT2, STAT1 and OAS1 were detected by qPCR. Data are means  $\pm$  s.d. from n = 3 independent experiments. **j**, Huh7, Huh7 GALNT2<sup>-/-</sup> and Huh7 STAT1<sup>-/-</sup> cells were treated with 100 ng/mL of IFN- $\beta$  for 24h. Then cells were infected with PR8 H1N1 at MOI of 0.01. Viral infection efficiency was detected the by IFA. Data are means  $\pm$  s.d. from n = 6 independent experiments. **k**, Schema depicting the presence and position of GAS and ISRE elements in the promoter region (2 kb upstream from the transcription initiation site) of human GALNT2 gene. All Western blots were representative of three independent biological repeats. Data are mean  $\pm$  s.d. *P* values from two-tailed Student's t-test. ((ns): no significant, \**P*  $\leq$  0.05, \*\**P*  $\leq$  0.01, \*\*\**P*  $\leq$  0.001, \*\*\*\**P*  $\leq$  0.0001).

**Extended Data Fig. 6. Increased lethality and pathogenicity of H1N1 PR8 in *GALNT2*<sup>-/-</sup> mice.**

**a**, Using CRISPR/Cas9-mediated gene editing technology, two single guide RNAs (sgRNAs) were designed targeting the second exon and the region between the second and third exons of the *GALNT2* gene. **b**, Sequencing analysis revealed the complete knockout of the second exon of *GALNT2*. **c**, WT (n=5) and *GALNT2*<sup>-/-</sup> (n=4) C57BL/6 mice were intranasally treated by 2 µg IFN- $\gamma$ . Lungs were harvested at 3 d.p.i.. *GALNT2* protein expression levels were determined by western blot. Western blots were representative of three independent biological repeats. **d**, WT (n=5) and *GALNT2*<sup>-/-</sup> C57BL/6 (n=5) mice were intranasally infected with 350 FFU of PR8 H1N1. Lungs were harvested at 3 d.p.i.. **e**, Quantitative PCR analysis of *Il-1 $\beta$*  mRNA in the lungs from **(d)**. **f**, HE staining of the infected lungs of WT (n=5) or *GALNT2*<sup>-/-</sup> (n=5) C57BL/6 mice infected with PR8 H1N1 at 3 d.p.i.. The scale bars are indicative of 100µm and 1000µm respectively. Data are mean  $\pm$  s.d. *P* values from two-tailed Student's t-test in **(b)**, **(d)**. ((ns): no significant, \**P*  $\leq$  0.05, \*\**P*  $\leq$  0.01, \*\*\**P*  $\leq$  0.001, \*\*\*\**P*  $\leq$  0.0001).

**Extended Data Fig. 7. Overexpression of GALNT2 alleviates pneumonia induced by IAV and SARS-CoV-2 infection.**

**a**, Groups of C57BL/6 mice (n=2) were either transduced with  $3.3 \times 10^{11}$  GC of AAV9-EV or AAV9-hGALNT2 intranasally. Lungs were harvested at 14 d.p.i.. GALNT2 protein expression levels were monitored by western blot. Western blots were representative of three independent biological repeats. **b-c**, AAV9-EV or AAV9-GALNT2 transduced C57BL/6 mice (n= 5 or 6) infected with 1000 FFU PR8 H1N1. Three days post infection, *GALNT2* expression levels (**b**) and PR8 M (**c**) gene mRNA in the lungs were analyzed by qPCR. Two of five mice in AAV9-GALNT2 group could not be detected by RT-qPCR (RT-qPCR Cq values above 38 were considered as negative). **d**, AAV9-EV or AAV9-GALNT2 transduced C57BL/6 mice infected with 1000 FFU PR8 H1N1. Three days post infection, lungs were harvested. Representative images of infected lungs were shown. **e**, HE staining of the lungs from (**d**). The scale bars are indicative of 1000 $\mu$ m and 500 $\mu$ m respectively. **f**, Quantitative PCR analysis of Il-1 $\beta$  mRNA in the lungs from (**b**). **g-h**, Groups of C57BL/6 mice were either transduced with  $3.3 \times 10^{11}$  GC of AAV9-EV or AAV9-GALNT2 (n =5 or 6) intranasally and subsequently transduced with  $2.5 \times 10^8$  FFU of Ad5-hACE2 after 9 days post AAV transduction. Five days after secondary transduction, mice were infected with  $1 \times 10^5$  FFU of SARS-CoV-2 (WT). GALNT2 and SARS-CoV-2 *N* genes mRNA expression levels in the lungs were determined by qPCR. RT-qPCR Cq values above 38 were considered as negative. **i**, Representative images of infected lungs from (**g-h**). **j**, HE staining of the lungs of from (**g-h**). The scale bars are indicative of 100 $\mu$ m and 1000 $\mu$ m respectively. **k**, Quantitative PCR analysis of Il-1 $\beta$  mRNA in the lungs from (**g-h**). Data are mean  $\pm$  s.d. *P* values from two-tailed Student's t-test. ((ns): no significant, \**P*  $\leq$  0.05, \*\**P*  $\leq$  0.01, \*\*\**P*  $\leq$  0.001, \*\*\*\**P*  $\leq$  0.0001).

**Extended Data Fig. 8. Overexpression of GALNT2 in A549 has a minimal impact on host genes.**

**a**, Scatter plot showing relative up- and down-regulated genes, identified by RNA-seq analysis of GALNT2-overexpressing A549 cells. GALNT2 expression was validated by western blots (Extended Data Fig. 4g). **b**, Proteomics: Volcano plot illustrates GALNT2-regulated proteins from the Stable Isotope Labeling By Amino Acids In Cell Culture (SILAC) experiment. Greater than 2-fold change in abundance upon GALNT2 overexpression and lower than 0.05 adjusted p value were set as the threshold criteria for GALNT2-regulated proteins (see method). **c**, Glycoproteomics: Volcano plot illustrates GALNT2-regulated O-glycosylation sites from the SILAC experiment. Greater than 2-fold increase in abundance upon GALNT2 overexpression and lower than 0.05 adjusted p value were set as the threshold criteria for GALNT2-regulated sites (see method). Pearson correlation coefficients and *p*-values by two-tailed test are indicated.

**Extended Data Fig. 9. GALNT2-dependent O-linked glycosylation at S810/813 of spike.**

**a**, Soluble GalNAc-T1, T2, T3, T4, T5, T7, T11 and T18 were overexpressed in FreeStyle<sup>TM</sup> HEK293T cells and purified by Ni-NTA chromatography. All proteins were analyzed by Western blots. Western blots were representative of three independent biological repeats. **b**, MALDI-TOF analysis of GalNAcylation reactions catalyzed by purified GalNAc-Ts on the synthetic spike S2' (TMPRSS2) peptides. Reactions were performed and analyzed as described in the Methods. An increase of 203 Da corresponds to addition of one GalNAc residue was only detected O-glycosylation by GALNT2. **c**, Soluble GALNT2 (WT and four mutations) were overexpressed in FreeStyle<sup>TM</sup> HEK293T cells and purified by Ni-NTA chromatography. All proteins were analyzed by Western blots. Western blots were representative of three independent biological repeats. **d**, MALDI-TOF analysis of GalNAcylation reactions catalyzed by purified GALNT2 on the synthetic spike S2' (TMPRSS2) peptides. Reactions were performed (3h) and analyzed as described in the Methods. An increase of 203 Da corresponds to addition of one GalNAc residue was obvious detected O-glycosylation by WT GALNT2, but an apparent decline in four mutants.

**Extended Data Fig. 10. The S810/813I of spike evades GALNT2 antiviral function in a SARS-CoV-2 replicon system.**

**a**, Constructing a reverse genetic complementary SARS-CoV-2 replicon system. The N gene is deleted from the viral genome and replaced with a GFP reporter gene. Through *in vitro* ligation and transcription, a SARS-CoV-2 RNA genome lacking the N gene but carrying the GFP reporter gene is obtained. Then RNA genome is electroporated into Caco2 cells stably expressing the viral N protein (Caco2-N, packaging cell line), allowing the production of SARS-CoV-2-N<sub>del</sub> replicon. Here we generated 2 replicons, comprising SARS-CoV-2 WT and S810/813I spike mutations. **b**, Fluorescent and bright field images of SARS-CoV-2 replicons, including WT or S810/813I. Images were representative of three independent biological repeats. Scale bars are indicative of 200  $\mu$ m. **c**, Caco2-N cells were infected by WT SARS-CoV-2 or S810/813I replicon at MOI of 0.01 for 24 or 48 h in presence of EV or GALNT2. Viral titers were determined by TCID<sub>50</sub>. Data are means  $\pm$  s.d. from n = 3 independent experiments. *P* values from two-tailed Student's t-test. **d**, Caco2-N cells were infected with SARS-CoV-2\_N<sub>del</sub>-WT and SARS-CoV-2\_N<sub>del</sub>-S810/813I viruses at MOI of 0.01 for 48h in presence of EV or GALNT2. Caco2-N cells were infected with SARS-CoV-2\_N<sub>del</sub>-WT and SARS-CoV-2\_N<sub>del</sub>-S810/813I viruses at MOI of 0.01 for 48h in presence of EV or GALNT2. The GFP-positive cells were quantified of syncytia among total cells in Caco2-N cells. Data are means  $\pm$  s.d. from n = 3 independent experiments. *P* values from two-tailed Student's t-test. Images were representative of three independent biological repeats. Scale bars are indicative of 200  $\mu$ m. **e**, A schematic shows the S810/813I and S810/813A of SARS-CoV-2 recombinant viruses evaded GALNT2 antiviral activity, while reducing the viral fitness due to reduced utilization of TMPRSS2 for viral entry.

470 **Supplementary tables 1 to 7**

471 Supplementary table 1. Information of SARS-CoV-2 infected participants.

472 Supplementary table 2. P values of the most significant SNP of genetic variation at the GALNT2 and Mx1 were  
473 displayed for H1N1 and H7N9 infections.

474 Supplementary table 3. Primers for RNA Quantification of human and mouse genes.

475 Supplementary table 4. Sequence alignment of S2' cleavage site sequences of coronaviruses spike.

476 Supplementary table 5. Sequence alignment of HA cleavage site sequences of influenza A viruses.

477 Supplementary table 6. Primers for generation of recombinant SARS-CoV-2 viruses.

478 Supplementary table 7. Primers for generation of recombinant SARS-CoV-2 replicon.

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482 **Supplementary table 1.** Information of SARS-CoV-2 infected participants.

| Patient_ID | Age (years) | Disease severity | Sex    | Tissue | n_Samples | Sampling time points |
|------------|-------------|------------------|--------|--------|-----------|----------------------|
| nCoV-3     | 64          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-8     | 61          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-13    | 79          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-7     | 53          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-11    | 58          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-19    | 41          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-6     | 49          | Severe           | Male   | PBMC   | 1         | T3                   |
| nCoV-12    | 26          | Severe           | Male   | PBMC   | 1         | T3                   |
| nCoV-30    | NA          | Severe           | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-B1    | 56          | Mild             | Female | PBMC   | 2         | T1,T2                |
| nCoV-16    | 50          | Mild             | Male   | PBMC   | 3         | T1,T2,T3             |
| nCoV-17    | 82          | Mild             | Female | PBMC   | 3         | T1,T2,T3             |
| nCoV-B13   | 16          | Mild             | Female | PBMC   | 3         | T1,T2,T3             |
| nCoV-15    | 25          | Mild             | Female | PBMC   | 3         | T1,T2,T3             |
| nCoV-Y2    | 57          | Mild             | Female | PBMC   | 3         | T1,T2,T3             |
| HC-01      | 42          | HC               | Female | PBMC   | 1         | T0                   |
| HC-02      | 47          | HC               | Female | PBMC   | 1         | T0                   |
| HC-03      | 56          | HC               | Male   | PBMC   | 1         | T0                   |
| HC-04      | 52          | HC               | Male   | PBMC   | 1         | T0                   |
| HC-05      | 62          | HC               | Male   | PBMC   | 1         | T0                   |
| HC-06      | 49          | HC               | Male   | PBMC   | 1         | T0                   |

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489 **Supplementary table 2.** *P* values of the most significant SNP of genetic variation at the  
490 GALNT2 and Mx1 were displayed for H1N1 and H7N9 infections.  
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| Gene   | Data Source              | Phenotype                 | No. of subjects |          | Most Significant SNP | Annotation | Reference/Alternative active allele | Alternative allele frequency |          | Risk allele | Odds ratio of the risk allele (95% confidence interval) | P value | GTEx eQTL association of the risk allele (whole blood) |
|--------|--------------------------|---------------------------|-----------------|----------|----------------------|------------|-------------------------------------|------------------------------|----------|-------------|---------------------------------------------------------|---------|--------------------------------------------------------|
|        |                          |                           | Cases           | Controls |                      |            |                                     | Cases                        | Controls |             |                                                         |         |                                                        |
| GALNT2 | Dominguez et al. 2015    | H1N1 infection vs control | 49              | 107      | rs16851270           | intronic   | G/A                                 | 19%                          | 7%       | A           | 1.89 (1.54-6.59)                                        | 0.001   | GALNT2 down-regulation                                 |
|        | Yongkun Chen et al. 2019 | H7N9 infection vs control | 217             | 116      | rs72758262           | intronic   | G/T                                 | 32%                          | 20%      | T           | 1.89 (1.29-2.76)                                        | 0.001   | GALNT2 down-regulation                                 |
| MX1    | Dominguez et al. 2015    | H1N1 infection vs control | 49              | 107      | rs468105             | intronic   | A/C                                 | 33%                          | 48%      | A           | 1.89 (1.14-3.13)                                        | 0.01    | MX1 down-regulation                                    |
|        | Yongkun Chen et al. 2019 | H7N9 infection vs control | 217             | 116      | rs467593             | intronic   | G/A                                 | 30%                          | 46%      | G           | 1.92 (1.39-2.70)                                        | 0.0001  | MX1 down-regulation                                    |

493 **Supplementary table 3.** Primers for RNA Quantification of human and mouse genes.  
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| Species | Gene           | Primer sequence                                           |
|---------|----------------|-----------------------------------------------------------|
| Human   | <i>β-ACTIN</i> | F: AGGCCAACCGCGAGAAGATGACC<br>R: GAAGTCCAGGGCGACGTAGCAC   |
|         | <i>MDA5</i>    | F: GCTGAAGTAGGAGTCAAAGCCC<br>R: CACTGTGGTAGCGATAAGCAG     |
|         | <i>cGAS</i>    | F: CAGCCGCGATGATATCTCCA<br>R: GCAGAAATCTTCACGTGCTCATA     |
|         | <i>RIG-I</i>   | F: CACCTCAGTTGCTGATGAAGGC<br>R: GTCAGAAGGAAGCACTTGCTACC   |
|         | <i>MAVS</i>    | F: ATGGTGCTCACCAAGGTGTCTG<br>R: TCTCAGAGCTGCTGTCTAGCCA    |
|         | <i>IRF3</i>    | F: TAAACGCAACCCTTCTTTGC<br>R: GATGCACAGCAGGAGGATT         |
|         | <i>IFN-α2b</i> | F: GCTTGGGATGAGACCCTCCTA<br>R: CCCACCCCTGTATCACAC         |
|         | <i>IFN-β</i>   | F: GAGCTACAACCTTGCTTGGATTG<br>R: CAAGCCTCCCATTCAATTGC     |
|         | <i>STAT1</i>   | F: ATCAGGCTCAGTCGGGGAATA<br>R: TGGTCTCGTGTCTCTGTTCT       |
|         | <i>MX1</i>     | F: GCAAATCAAGGCACTGGAAG<br>R: TCGAAACATCTGTGAAAGCAAG      |
|         | <i>OAS1</i>    | F: TGTCCAAGGTGGTAAAGGGTG<br>R: CCGGCGATTAACTGATCCTG       |
|         | <i>GALNT2</i>  | F: CCGCAACAAGTTCAACCAGG<br>R: ATGGGGCGGGCTTTTCTTAA        |
|         | <i>ACE2</i>    | F: ATCAGAGATCGGAAGAAGAAAA<br>R: TCATCAGTGTTTTGGAATCCTGG   |
| Mouse   | <i>Gapdh</i>   | F: TGATGGGTGTGAACCACGAG<br>R: AGTGATGGCATGGACTGTGG        |
|         | <i>Il-1b</i>   | F: ACTGTTTCTAATGCCTTCCC<br>R: ATGGTTTCTTGAGACCCTGA        |
|         | <i>Il-6</i>    | F: GAGGATACCACTCCCAACAGACC<br>R: AAGTGCATCATCGTTGTTCATACA |
|         | <i>Tnf</i>     | F: GAACTGGCAGAAGAGGCACT<br>R: AGGGTCTGGGCCATAGAAGT        |
|         | <i>Ccl2</i>    | F: CTTCTGGGCCTGCTGTTCA<br>R: CCAGCCTACTCATTGGGATCA        |
|         | <i>Cxcl10</i>  | F: GCCGTCATTTTCTGCCTCAT<br>R: GCTTCCCTATGGCCCTCATT        |
|         | <i>Ifn-α4</i>  | F: TCCATCAGCAGCTCAATGAC<br>R: AGGAAGAGAGGGCTCTCCAG        |
|         | <i>Ifn-β</i>   | F: TCAGAATGAGTGGTGGTTGC<br>R: GACCTTTCAAATGCAGTAGATTCA    |
|         | <i>Stat1</i>   | F: ACAACATGCTGGTGACAGAGCC<br>R: TGAACAACTGCCAACTCAACACCTC |

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**Supplementary table 4.** Sequence alignment of S2' cleavage site sequences of coronaviruses spike. Labeled S and T represent potential sites that could be O-glycosylated by GALNT2.

| Virus                | Sequence                                               | GenBank sequence |
|----------------------|--------------------------------------------------------|------------------|
| SARS-CoV-2 (Wuhan)   | 809P <b>S</b> KP <b>S</b> KR↓SFIEDL821                 | NC_045512.2      |
| SARS-CoV-2 (Alpha)   | 768P <b>S</b> KP <b>S</b> KR↓SFIEDL780                 | OM189517.1       |
| SARS-CoV-2(Beta)     | 806P <b>S</b> KP <b>S</b> KR↓SFIEDL818                 | MZ433432.1       |
| SARS-CoV-2 (Delta)   | 809P <b>S</b> KP <b>S</b> KR↓SFIEDL821                 | OK091006.1       |
| SARS-CoV-2 (Omicron) | 804P <b>S</b> KP <b>S</b> KR↓SFIEDL816                 | OP093374.1       |
| SARS-CoV             | 791PLKP <b>T</b> KR↓SFIEDL803                          | NC_004718.3      |
| MERS-CoV             | 881TG <b>S</b> <b>S</b> AR↓SAIEDL893                   | MH454272.1       |
| HCoV-OC43            | 902C <b>S</b> K <b>A</b> <b>S</b> <b>S</b> R↓SAIEDL914 | KF963244.1       |
| HCoV-229E            | 681G <b>S</b> RVAGR↓SAIEDI693                          | KU291448.1       |
| HCoV-NL63            | 864 <b>S</b> <b>S</b> RIAGR↓SAIEDL876                  | KM055650.1       |
| HCoV-HKU1            | 894QCG <b>S</b> <b>S</b> <b>S</b> <b>S</b> R↓SLLEDL906 | MH940245.1       |
| Bat-CoV-RaTG13       | 805P <b>S</b> KP <b>S</b> KR↓SFIEDL817                 | MN996532.2       |
| Pangolin-CoV-GX_P2V  | 805P <b>S</b> KP <b>S</b> KR↓SFIEDL817                 | MW532698.1       |
| SADS-CoV-CH-JX1      | 667ANKYE <b>S</b> R↓SAIEDL679                          | OK078018.1       |
| Canine-CoV           | 973PPRIGKR↓SAVEDL985                                   | OK340213.1       |
| MHV-A59              | 863P <b>S</b> AIRGR↓SAIEDL875                          | MF618252.1       |
| Feline-CoV           | 750P <b>T</b> <b>T</b> IGKR↓SAVEDL762                  | KX722530.1       |

501 **Supplementary table 5.** Sequence alignment of HA cleavage site sequences of influenza A  
502 viruses. Labeled S and T represent potential sites that could be O-glycosylated by GALNT2.

| Virus         | Sequence              | GenBank Sequence |
|---------------|-----------------------|------------------|
| PR8(H1N1)     | 338IPSIQSR↓GLFGAI350  | OM486959.1       |
| H2N3          | 334VPQIESR↓GLFGAI346  | MT624542.1       |
| H2N5          | 334VPQIESR↓GLFGAI346  | MH637193.2       |
| H3N2          | 339VPEKQTR↓GIFGAI351  | DQ415325.1       |
| H4N6          | 337IPEKATR↓GLFGAI349  | AF285885.2       |
| H5N1          | 340ERRRKRR↓GLFGAI352  | AY854190.1       |
| H6N2          | 338VPQIETR↓GLFGAI350  | AJ410541.1       |
| H7N9          | 337KRKRTR↓GLFGAI349   | KY751058.1       |
| H8N4          | 338TPSIEPK↓GLFGAI350  | CY140088.1       |
| H9N2          | 332VPAASGR↓GLFGAI344  | AY790313.1       |
| H10N7         | 334PELMQGR↓GLFGAI346  | KY688101.1       |
| H11N1         | 336VPTIATR↓GLFGAI348  | CY146093.1       |
| H12N2         | 336VPQAQNR↓GLFGAI348  | CY190019.1       |
| H13N2         | 337VPAISNR↓GLFGAI349  | CY239408.1       |
| H14N3         | 341IPDKQAK↓GLFGAI353  | KX602668.1       |
| H15N9         | 343PEKIRTR↓GLFGAI355  | CY130102.1       |
| H16N3         | 336VPSSINGR↓GLFGAI348 | CY177441.1       |
| H17N10        | 337PQMEGR↓GLFGAI348   | CY103892.1       |
| H18N11        | 333NPIKETR↓GLFGAI345  | KR077932.1       |
| IBV(Victoria) | 356AKLLKER↓GFFGAI368  | M58428.1         |
| IBV(Yamagata) | 354AKLLKER↓GFFGAI366  | CY018765.1       |
| ICV           | 440VTKPKSRIFGIDD452   | AM410041.1       |
| IDV           | 449TLTPATRIFGIDD461   | OM468245.1       |

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**Supplementary table 6.** Primers for generation of recombinant SARS-CoV-2 viruses.

| Species         | Gene          | Primer sequence                                                                            |
|-----------------|---------------|--------------------------------------------------------------------------------------------|
| Authentic virus | S810/8<br>13A | F: ggttttaattttcacaaatattaccagatccaataaaaaccaatcaagaggtcatttataggatgacgacgataagtaggg       |
|                 |               | R: tgttgaaaagtagatcttcaataaatgacctcttgattggttttattggatctggtaatagccagtggtacaaccaattaacc     |
|                 | S810/8<br>13I | F: tggtttaattttcacaaatattaccagatccagcaaaaccagccaagaggtcatttataggatgacgacgataagtaggg        |
|                 |               | R: TGTGAAAAAGTAGATCTTCAATAAATGACCTCTTGGCTGGTTTTGCTGGATCTGGTA<br>ATAGCCAGTGTTACAACCAATTAACC |

**Supplementary table 7.** Primers for generation of recombinant SARS-CoV-2 replicon.

| Primer  | Sequence (5'-3')                                                                           | Usage                                              |
|---------|--------------------------------------------------------------------------------------------|----------------------------------------------------|
| THU7069 | taatacgactcactatagattaaaggtttataccttcccag                                                  | Fragment 1                                         |
| THU7070 | tgaagccaatttaaactcaccagactcatcaaat                                                         |                                                    |
| THU7071 | gcaccaacaaaggttacttttggtgatg                                                               | Fragment 2                                         |
| THU7072 | accacccttaagtgtctatcttgggttac                                                              |                                                    |
| THU7073 | gaagttactggcgatagttgaataactatatgctc                                                        | Fragment 3                                         |
| THU7074 | ttcaaaggcttcagtagtatcttagctaagagaatg                                                       |                                                    |
| THU7075 | gtttatgattacttagttctacacaggagtttag                                                         | Fragment 4                                         |
| THU7521 | gggtaaccattaactgataattcattttaaaaccatc                                                      |                                                    |
| THU7308 | aaagtaggcatacttgcataatgtctgatagagaccttatgacaagtgcaatttac                                   | Fragment 5                                         |
| THU7309 | GCCGTTAACTTTTGTGCACAAATGAGGTCttagcagcaatacaccaaggcaatcac                                   |                                                    |
| THU7310 | ggtgattgccttgggtgatattgctgctagaGACCTCATTTGTGCACAAAAGTTTAACGG                               | Fragment 6                                         |
| THU7286 | TATGTAGCTTTTCGACATGTTgcggccgcggatagttcctccttcagcaaaaaacc                                   |                                                    |
| THU7285 | gggggttttctgtaaggaggaactatatccgcggcgcAACATGTGCGAAAGCTACATATAAGGAACGT                       | Fragment 7                                         |
| THU7090 | GCTTACAGACAAGCTGTGACCG                                                                     |                                                    |
| THU7068 | aagttggttggttggttacctgggaaggtataaacctttaatctatagtgagtcgtattaAACCCACTATTTATACCAT<br>GGGAGGC | Fragment 8                                         |
| THU7089 | GTGATGACGGTGAAAACCTCTGAC                                                                   |                                                    |
| THU2140 | TACTGTAATACGACTCACTATAGGATGTCTGATAATGGACCCCAAATC                                           | T7-N                                               |
| THU2141 | TTTTTTTTTTTTTTTTTTTTTTTTTTTTTTGGGAGCTCGCCCGGGGATCCTTAGGCCTG<br>AGTTGAGTCAGCAC              |                                                    |
| THU7607 | cagatccaATTaaaccaATAaagaggtcatttattgaagatctactttcaacaaag                                   | For introduce S810I<br>and S813I to Fragm<br>ent 5 |
| THU7608 | atgacctctTATtggtttAATtgatctggtataattgtgaaaaataaaaccaccaaaatcttaattggtg                     |                                                    |